SI412: Portable Alpha: Asking the Questions That Matter ft. Harry Moore

Podcast: Top Traders Unplugged

AI Executive Summary

1. Core Thesis The episode argues that investment success relies on resilience and preparation for unpredictable market events rather than prediction, utilizing a rules-based approach to manage risk exposure across asset classes. Through interviews with Harry Moore from Man Group, the discussion centers on how AI is transforming quantitative research productivity by automating "table stakes" coding tasks in favor of human creativity; how trend following strategies can serve as vital diversification during left-tail shocks through cash-efficient structures like portable alpha and volatility-based scaling (Kelliness criteria); and why historical risk management principles found in the Turtle Trading methodology remain relevant when linked to modern decision theory.

2. Key Arguments & Data

3. **Market Environment & Trend Performance:** The market trend barometer reads 27, indicating a weak signal for softness at the beginning of August. July saw significant dispersion among managers driven by themes like the AI rally stopper (which hurt energy and tech sectors) versus rising equities in traditional markets; slow-moving models performed better as they remained long stocks through volatility cycles while maintaining persistent short positions on bonds and flexible exposure to commodities/energy which rose year-to-date.
4. **AI & Quantitative Research:** Harry Moore posits that AI tools have turned Python coding from a creative necessity into "table stakes," allowing researchers to focus 90% of their time on strategic thinking rather than implementation details (the remaining 10%). Consequently, the best talent will not necessarily be top coders but those capable of rigorous critical analysis and creativity using AI. The industry faces a hiring challenge in verifying the quality of these new generations who may lack deep programming knowledge due to generational differences and LLM usage.
5. **Portable Alpha Structures:** A significant paper was released ("Ask the Hard Questions") focusing on robust construction strategies:

- **Cash Buffers:** To avoid margin call liquidity risks during stress events (GFC, Dot-com bust, COVID), structures require roughly 40% cash support for beta allocation (10% as initial margin) to survive drawdowns without forced liquidation. A lower buffer significantly increases the probability of being "knocked out" in large losses over a 26-year timeframe.
- **Rebalancing Rules:** Monthly rebalancing improves tracking error versus an index during bear markets (e.g., 2008), but annual or longer holding periods are theoretically preferred for long-term performance and reduced volatility drag on the beta component. Rebalancing triggers if alpha/beta drifts >10% out of whack, though buy-and-hold is viable given equities' historical reversion-to-mean over long horizons.
- **Futures vs Swaps:** Futures are operationally preferred due to counterparty risk avoidance and liquidity availability, even with slightly higher implicit costs in swaps; the "no free lunch" thesis holds true across both vehicles structurally if operational constraints allow.

6. **Turtle Trading Mathematics:** A paper by Paulson & Circle links 1980s Turtles rule-of-thumb calculations (volatility estimation via high-low ranges and exponential decay) to modern decision theory:
   - **Position Sizing (N):** The original N volatility estimator is an efficient method for estimating risk; this directly connects to the Kelly Criterion, where optimal position sizing balances expected return against ruin probability. Turtles historically set stops at 2% of total equity (a prudent version of Kelly) rather than risking more on potential upside growth without sufficient safety margin.
   - **Correlation Management:** Instead of inverting covariance matrices which is complex for historical manual calculation, correlated markets were managed via position size caps or limits to prevent correlation blow-ups during stress periods.

7. **Leverage & Diversification Concerns:** Investors often view stacking return streams (Beta + Alpha) as "leverage risk." The counter-argument posits this is not true leverage but portfolio construction for optimized Sharpe ratio and cash efficiency; since trend strategies scale down positions based on volatility during crashes, they do not

exacerbate tail risks like short-volatility or liquidity provision strategies which often increase correlation during stress.

8. Actionable Takeaways

9. **Optimize Portfolio Construction:** For those utilizing portable alpha structures (e.g., futures vs swaps), implement a cash buffer strategy that allows for margin calls without forcing liquidations of the beta component, ensuring unencumbered capital remains available through major market shocks. Consider using longer lookback periods in trend models to capture persistent macro trends over short-term noise and manage volatility risk better than high-frequency systems this year.
10. **Adapt Hiring & Research Processes:** When employing AI-assisted quant teams or researchers, prioritize testing candidates' ability to critically evaluate and modify AI-generated code rather than assessing raw coding proficiency alone. Focus hiring on strategic creativity while accepting that the "table stakes" of implementation will be automated by tools.
11. **Understand Structural Risks:** Review your portfolio's covariance matrix specifically for left-tail correlations during stress events (e.g., equity drawdowns). Ensure alpha components are uncorrelated with beta not just in normal times but under adverse market conditions, and verify that position sizing scales inversely to volatility as predicted by Kelly criteria.
12. **Monitor Market Universe Divergence:** Be aware of performance dispersion between traditional futures markets (e.g., US Treasuries) versus older/OTC instruments (e.g., Polish interest rate swaps or cross-sectional factors). If trend models diverge significantly, it may indicate a shift in market regime favoring long equities and short bonds.

Full Transcript

Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing, economics, and beyond, to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence. Welcome and welcome back to this week's edition of the Systematic Investors series with Harry Moore and I, Niels Kastroblasen, where each week we take the pulse of the global markets through the lens of a rules-based investor. I also want to say a warm welcome. If today is the first time you're joining us and if someone who cares about you and your portfolio recommended that you tune into the podcast, I would like to say a big thank you for sharing this episode with your friends and your colleagues. It really does help us. Anyways, Harry, it's wonderful to see you and have you back on the show this week. How are you doing? What's going on in your part of the world? Yeah, great to see you, Niels. All good over here in New York. I think we've hit 100% humidity today. So a bit uncomfortable. A bit of an uncomfortable.

journey in but other than that all good over here and yeah thanks for having me back really appreciate it. Yeah no no absolutely we have a great lineup not least thanks to you and so i'm excited about it you know before we dive into all of that i love to just get a sense of kind of what's come across your desk that you found interesting doesn't have to be anything related to what we do but is there anything that you have come across that you have enjoyed recently? Yeah, I think, as I mentioned, I moved to New York kind of two and a half years ago and obviously you look around, you see the bridges, the highways like where this kind of all come from. So the book that is supposed to explain that best is the biography on Robert Moses as it's called The Power Broker. So I ordered this book, it's like 1300 pages. So I started kind of lugging that around on the subway.

It's a bit of a brick, to be honest. So I've kind of given up. I don't know. It's a failure on my part, but I've given up and I've gone for the Kindle. So I've been reading about it, but it's just very interesting just to see, like, why is the west side a huge road when we could have had some coastline here? But you just get a feel for it. It's a bit of a biography of New York in a way. I feel for how the city's been built. So been making my way through that, haven't quite finished it yet. But yeah, did give into the paperback and had to go to Kindle on that, sadly.

Yeah, super interesting super interesting and now For my part by the way speaking of bricks. I don't know if you remember but You know back in the very early days of mobile phones Actually my brother-in-law still keeps his first ever mobile phone and literally it was like a brick with an antenna on I mean it looks so weird now, but it's that's how people were carrying them around back then Amazing. So my first was a 3210. I don't know if you remember the 3210, but it had had snake in it. I think it did actually have a tiny little antenna, but it was one of the first ones that you could like easily kind of carry around in your pocket. So not quite as far back as those huge brick phones. But yeah, this book would have put me in good stead for lugging around one of those old phones, I think Niels. Yeah.

So on my radar, nothing really surprising, I guess, but it's just some of the headlines that I followed. Nothing really that is, well, somewhat linked to what we do, of course. But the first one is this story about this hedge fund young guy, Leopold Aschenbrenner, and the situational awareness. And of course, you read the headlines and the news stories and you probably shouldn't believe everything that that it says but in any event i mean some of the things that you do read. The level of experience the level of assets that he suddenly commanded the level of leverage he was using that people were you know probably should have been aware of so to speak. There's just so many red flags in my opinion when you see that and yet for for many firms with you know.

decades of experience and having demonstrated their ability to manage risk and so on and so forth. You see how difficult it is to raise money and hear someone with very little experience, very little track record ends up having 45 billion dollars or so on the management before losing two thirds of that

it seems in a matter of weeks. I mean, it's closer probably from location wise to where you are. But it's, yeah, it's just a lot of red flags when you see something like that. Is that anything you talk about in New York at the moment? I imagine so. Yeah, I mean, I think it's a good example of where you can be right and still hit issues.

He'd called the AR trade. Obviously he had a great story from writing the essay. He was very well known in those circles. And while it was, I guess, known as a hedge fund, the hedge was essentially the other side of the AR trade. So even if on paper you look somewhat like dollar neutral or market neutral, your theme is. It's kind of caught you out there. I think you chatted through this a little bit with Yoav and Nick a few weeks ago on capturing those themes. And it was a big bet on a theme, and that theme obviously reversed in July, and that can show if you kind of compound on one idea what can happen if it goes wrong. Yeah, I think that's a good point, actually. That diversification really does matter, so to speak. The other thing...

Where i guess you could say it's not necessary that the people who act are always right. But when they do act it does have an effect in the markets and that's. The central banks of japan and possibly i guess the u.s as well intervening in the japanese yen only a couple of days before month and which. You know it did have an effect it did have a show up in i think in performance of our industry as well and but of course on the back of. 30 years of zero interest rates only a couple of years ago. The BDA started to raise rates for the first time in in ages and of course interest rates has continued to move higher despite and and the end has despite actually continue to move lower and but then they made a real effort at least to change that they've done it I think a couple of times.

before in the last two three years it's never really lasted very long but it was interesting that they decided to come out. Yeah it's one of those that from a trend perspective is tough as well right because sometimes I think it kind of works because you take that uncomfortable. Door position and then you hold it and you write it down and there's rumors is there gonna be intervention humans that's a panic the rules based systems don't that. Of course, on this occasion, the intervention, you did see a bum part. I mean, I did enjoy the photo. I wasn't sure if it was a meme. Apparently, it was real

of Scott Besson's to-do list for the day, and it just was a blank sheet of paper with one bullet point that said, bye, yeah. So I mean, I might need to backtrack if that was an AI-generated image, but it did make me chuckle.

But yeah, it's the importance of stizing and making sure that one position can't hit you too much. But yeah, a bit painful when an intervention like that makes something turn around. Sure. I mean, you probably know this much better than I do. But if I'm not mistaken, I think it was like a tin standard deviation move, which is rare. I know that when the Swiss franc got depict from the Euro, I think it was like 20 or 30 standard deviations or something like that. But still, 10 standard deviation is not something we should expect really to see. Not at all. Not at all. That does not appear. And if you're building market data, it's very hard to actually simulate something like that. But we know they exist from time to time. I think when you've had a move go so far for so long, natural market forces weren't bringing it back. And I think that was where they decided to intervene and that human interventions caused the, as you say, very, very large move in the end. Yeah. All right, well, let's do a little bit of trend following update and then we'll get into some of these wonderful topics that we're going to be talking about. So my trend barometer finished yesterday at 27. That's a very low reading, actually.

It's a pretty weak signal. So it suggests some softness to the beginning of August, which we'll see also in the numbers in a second. We just finished July, which was I would say a month that I mean it's not there was overall that bad, but there was definitely some dispersion among managers, which of course there always is, but still maybe more than I would normally expect in a single calendar month. Any. Any observations, any takeaways from your side so far, whether it's August, July, whether it's year-to-date, anything that you find interesting about the environment for trend following at the moment? Yeah, I mean, July was, if you look at July, it was kind of characterized by your sort of two big themes. I would say you had a bit of a stopper on the AI trade as we already briefly discussed with obviously one hedge fund being, we're taking some.

pretty big losses. You also, of course, had rallying energy markets again. I think from a trend model perspective, if you're taking sensitivity around

three months, like a continuous moving average, it probably would have been long markets like the oil complex or it would have been long stocks. I think this is why you're seeing such dispersion. So choices about how much to have within each asset class, how much risk weight to have in the long run, choices of whether to exclude certain markets or whole asset classes would have created a lot, a lot of dispersion in July. So I would expect people long equities would have been good. That would have been, sorry, tough in general, particularly if you were exposed to semis.

those trend following kind of themes as well. So if you were trend following themes and you were long, let's say, tech hardware companies, you're long semiconductors and you were short kind of software like that would have hurt as well. So I think that kind of explains a bit of that dispersion you're seeing there. There was more energies, would have been good, more stocks, generally bad. So yeah, big, big dispersion in July. I don't know if you managed to listen to it, but last week I had Rich and Dave Dredge on the show and there was one thing that really, it kind of, it wasn't surprising in that sense, but still when you hear it, you're kind of, you think, wow. And that was this example that Dave brought up early on in the conversation about I think it's the South Korean stock, SK Hynex or something like that. Where the stock had gone up like 120% and then down to flat for over a three month period. But the 2X levered ETF was down like 40 plus percent for that period. Again, obviously highlighting the dangers of leverage and how some of these, quote unquote, retail products are.

i put together but anyways suddenly has been an interesting summer so far but as i said i think for a lot of people is really the end that kind of defined whether you are gonna you know how you're gonna finish the month of july so let's look at that so the beat up 50 obviously we did the july figures last month but as i said last week but as i said actually it's. little bit soft getting into August. So we are down 67 basis points for the BTOP 50 and that would be as of Tuesday night. We're recording on Thursday this week. It's still up 7.17% so far this year. The SOC GenCG index is down 88 basis points up 7.15% so far this year. The trend index down about a percent or so 94 basis points up 6.89% for the year.

and the short-term traders index actually up for the month 14 basis points but up just 3.32% so far this year. If we look at the traditional markets we have the MSCI world index and this is as of yesterday up 2.77% in August up 12.45 for the year. Stripping out US and Canada based on the proposal from a listener. That index, so the EVE index is up 1.83% in August down, sorry, up 11.8% for the year. US S&P-US aggregate bond index up 64 basis points in August, up 31 basis points this year. And the S&P 500 total return up 3.12% so far this month and up 12.83% this year. really strong, by the way, start this month to equities for sure. I mean, from my perspective before we move on, not a massive amount of change, directional change in kind of a trend following portfolio certainly changes in terms of the levels of commitment. But generally speaking, traders are still long equities probably long metals probably as you said long energies. Probably long the dollar I would imagine and then still most likely short fixed income to some extent so not a lot there when I look at another thing that you can comment on if you want don't have to when I look at like a like a standard trend following model and I compare just different.

speeds because you mentioned speed could have been a determinant. And I think that that's very true. So I tend to look at a traditional trend following model and looking at say four different speeds, meaning look back periods, 20 days, 60 days, 130 days, 260 days. And if you had chosen a 20 day look back again, this is just from this specific model, you would actually have been down this year. And if you had chosen a 60 day, you would also have been down for not as much. If you had chosen 130 days, you would have been slightly up. And if you had chosen an even further out 260 days, you would have been up a little bit more. That's kind of how I see speed impacting at the moment, the environment or the results. I don't know if you have anything that you want to say, comment on here. Yeah, I would say that.

That completely makes intuitive sense to me. So if you're running a slower trend system in 2026, you would have been fairly more persistently short bonds, which I would have certainly paid in the past few months. Once you built into those long energy positions, you would have tended to hold them a little bit longer, which on a year-to-date basis. would have helped. You might have suffered a little bit more volatility there, but would be also

adding. And then in stocks, I mean, the longest models have done the best there as well. So very long look backs. You've been persistently long stocks. Again, we've had some wobbles along the way, but if you look at stock indices in general across the board, they're up. So it's certainly been a year so far, I would say like slower has been beneficial.

We haven't had a turning point that's persisted for long enough for the fastest beats to kind of shine. So, Echo is a little bit of what we've seen over the past few years that SLO has tended to outperform a bit. Again, we've done research on this for you've discussed it on your show. I mean, my view is always to have some dynamism there that you draw. you can still capture some of those turning points. Well, but yeah, I completely agree with what you said in general, as you've gone longer out eventually, you've held on to your long energy positions, you've tended to hold on to your short bond positions, and you've just been long stocks throughout the year, which has worked well. One thing I don't personally follow so closely, I would imagine you do more so, and that's because at Done, we just trade kind of the most liquid futures, but When with the other thing that drives performance as we often talk about before we get into the topics is of course market universe you've written. You know beautiful paper about this that was published in late January actually so I don't know if you remember this is not something we kind of plan to talk about but do you think there's been a lot of dispersion based on market universe this year as well. Yeah absolutely.

I would say this this year, you've really seen your kind of old versus traditional markets, so like diverged to a fairly large extent. I think it's two places where this is most evident, or maybe three actually. So the three that we could touch on would be in alternative energies. in fixed income and then also in stock. So in energies, if you go down that kind of old-fashioned market spectrum, you're starting to look at things like Nordic power or Italian power. Those markets have just tended to be a little bit more range bound than we've seen in the more traditional energy markets this year. So again, mid-frequency models tending to be profitable year today in things like brood.

eating oil, etc., and less so in some of the old energy use. On fixed income, like fixed income has been pretty difficult for a few years for trend now, right? Like, as you know, you've had these kind of competing forces in fixed income. You've got the administration shouting for red cards, red cards. You have markets follow that occasionally. Then you look at the inflation branch, you look at rising deficits, and you've got these kind of two forces. And I think that's been even more pronounced as you move down the spectrum and away from your big, traditional government bond futures, away from 10-year treasury and 10-year guild and you go and look at, for example, Polish interest rates swaps or other markets like this. Again, those kind of gyrations have just been more pronounced and they tended to be a little bit more difficult. The final one to talk on, We did discuss this to an extent in the paper. And actually, my colleagues have recently written a paper on this as well. If we get time to talk about it today, great. But otherwise, we can, of course, save it for another time. Save it for next time, yeah. But essentially there, typically, when you move away from your stock indices, so move away from just like SAP and NASDAQ, Euro stocks, and you start to look at cash equity sectors, for example. So you might look at, for example, like STEMIs versus software, as we mentioned in the intro, you might also look at things like value, momentum, quality. Those are much more cross-sectional constructions. So as we were discussing, a slow model would have been long the indices. The indices have gone up this year. That's worked well. These cross-sectional models, I mean, from a high level, have also been profitable just in a very, very different way.

Right, so if you think if you construct a quality factor, quality has really struggled this year on a year-to-date basis, each door of that factor, you can imagine may have added as well. So you've got three areas, I think, which have contributed like a big portion of the kind of variance between what you might call your traditional futures and forwards versus your more kind of OTC type instrument. So yeah, great, so great note. Tonight is that news and I think that's absolutely worth a comment. All right, well, let's move on to the topics you brought along. And before we get into kind of the papers, so to speak, I think you very nicely laid out something that ties into the initial part of our conversation, both in terms of AI and macro. So why don't you take us into today's main feature, so to speak?

Yeah, absolutely. Thanks, Niels. I think this was motivated by a few things. Of course, the events last in the past couple of weeks with the unwind and people stepping in to buy this AI trade that hit the headlines, but also really motivated by your conversations you've been having on diversification. David Dredge and Richard Brennan last week and then a couple of weeks back with Nick and with Yoav. Like from a trend following perspective, like looking at the AI trade and AI theme in markets, it's a very, very interesting time to kind of be part of it and be living through it, right? Because on the one hand, you have all of these comparisons back to the railroads, back to laying cable in the dot-com bust and these forecasts that it has to be a bubble, right? This is one side of the oven. It has to be a bubble. Like it's got all the echoes, it's got all the hallmarks.

of these prior periods where CapEx went through the roof. It's kind of manier about a new thing that's appeared and it all kind of all ends in tears, so to say. Not to say that there isn't some very useful infrastructure that's left behind, but from a market's perspective, it's very painful. That's kind of one side of the argument. The thing is, I always try and really stay balanced. That is a gripping narrative. It's a great story. like it's a gripping narrative to kind of tell to a client in a meeting. But you've also got to consider the other scenario, the kind of upside scenario. And like if I look within man group, we are integrating AI across all of our teams, across all of our processes. We're seeing benefits to quant research, like just huge amount of productivity gains is like, what do you want your researchers to be doing? You mostly want them to be thinking and being creative rather than writing lots and lots of Python code.

where the Python code is now kind of table stakes and you can do it much, much faster with these tools. Do I see the benefits here? And you hear obviously the frontier labs as they would say, talking about what the future might look like. And so I always try and stay balanced on this. Maybe there is this huge upside scenario where the trade that we saw go wrong last month is the right trade and it works over the next 10 years. But there's this kind of real level of uncertainty. And I think it links to your kind of tagline of the show, right? Like being prepared for what you just really can't predict. And that's how I think kind of trend following comes into it. Because if you think of, you're trying to build a portfolio to be resilient in

this environment, right? So you're allocating to many different asset classes, probably in an allocated portfolio have a decent amount of equities.

and have a decent amount of bonds, I probably hold potentially some diversifiers like commodities, you might hold some private assets. Now, in that portfolio, AI trade keeps going, economic growth keeps rising, CAPEX keeps rising, that portfolio is rising. Now, on paper, you look diversified, right? These assets, they seem to have in normal times somewhat low correlations with each other. You feel diversified, you feel kind of comfortable. Now, in a crack to that trade or a full blow up. And again, I mean here, like the S&P isn't down five or 10, it's down 30 or 40% over a period. When you look at that book that you own, like what in there is going to protect you against that huge drop? So you would say, well, the equities we've just said, potentially they're down 30 or 40.

Private assets, you can slow your marks. They may mark down more slowly, but at some point you're going to take hits there, right? And we've always seen the slowdown of return of capital. We've seen like the FT and others running articles on what a client's going to do when they're getting a lot less capital back than they expected. You're certainly not getting any long-term protection there. You might get some paper protection for a period. Then in bonds and gold, like earlier in the year, we saw, again, both of those sell off in a fairly correlated manner. The difficulty as well is if you look at the environment today and you look at the Fed in the US and you say, well, inflation has now been running above target for many, many years. It's not just like it's a small period, it's many years. Do they have the tools to really cut rates, like we've seen in the past, to kind of boy market?

If you're worried, like maybe they do, but if you're slightly worried about that, then you might be in an environment where your bonds are also selling off very, very hard as well. And this is where I think trend kind of comes into the equation because as we know, trend doesn't have a long-term view on any of these markets, right? It trades them all. Okay. It doesn't trade private assets, but we're trading entities, you're trading bonds, gold, oil, et cetera. It doesn't have a long-term view on those. It's just simply following the prices and saying, well, what I'm seeing in the last two weeks, in the last four weeks, in the last two months is actually the narrative is changing. And

I don't actually want to be long bonds and long equities. I actually want to be short bonds and short equities. If it's an inflation shock, you want to be long commodities. And so, Brent actually moves into those positions. Now, that's not to say it's always going to get it right.

It's going to cause you some pain from time to time. It's really being prepared for this left tail. The scenario I described at the start, when there is a crack, there's a break, maybe AI does deliver, but it's just not at the level that people are priced in. Having an asset like TrendFollowing in your book that can go short bonds, it can go short equities, as we described, it might even go short the AI fee. It might start to short semis and short the tech hardware builders. You could go short those names. Really, it's just offering that something completely different in your book that as an investor, when you look and you say, okay, my assets are uncorrelated, but are they uncorrelated in that left-tail shock? What does your covariance matrix look like?

In that shock and typically it's it's trend following that is that is zero or kind of negatively correlated and other asset classes of correlations typically typically pick up. Yeah, yeah, I mean you're kind of tempted. To say that the beauty of trend following is that it's aware of the situation it's situational aware but I think that name is taken unfortunately or maybe fortunately in this case. Now let me ask you something, you mentioned something that because I'm not a quant I find it interesting. You mentioned this thing about the productivity gains that you see from having AI tools and so on and so forth. Just sort of on a very basic level. I understand. The gains that someone like yourself who were taught how to program you can essentially tell it what to do and you can also review.

the output. Are you, when you sit in your quiet moment in the underground, on your way home, are you thinking, well, what about the next generation where they will use AI as kind of their first tool, but they actually don't have the background and experience in having done programming myself. So how are they going to tell if what the output was is any good? Is that something you think about at all? Yeah, absolutely, Niels. And I think it's the best answer there. So I go back and kind of the story of when I joined man group. So I'm originally an actor and basically in an act where I

worked 10, 15 years ago, programming is not a big part of the syllabus, right? So when I joined man, the first thing I did was go ahead first into all the develop courses run here and spend hundreds of hours in Python, being able to do the financial analysis, the sort of simple pandas, data manipulation, et cetera. Now, today, when I look at the tools available, like that was essentially a complete waste of time. So to the level that I can program, It's absolutely complete useless waste of 200 hours, sort of seven years ago now. At that level, the tools are absolutely fantastic. Now, if you think of a quant researcher and what you're really employing them for, what's the value of them in their seed?

five, six, seven years ago, how's your time spent? Well, it might be 50% doing the creative stuff, thinking, and 50%, maybe more than 50% is on implementation details, right? Being good at Python, knowing how to run and set up a back test, maybe replicating a test across multiple assets. Now, if you think where the value is, the value is not really in that second 50%, right? There's many, many more people. maybe five, maybe 10 to one times as many, who you can employ to do the second 50%. It's kind of table stakes in quantitative research, like being good at Python, being able to run a back test. The way the value really, really is, is that if you could blip that 50-50 and make it more like 90-10, so that your top researchers have plenty more time to think and be creative, I think that's a big, big AI enabler.

AI, well, it can summarize everything that's been written before, it can summarize the tax code. Very, very difficult to come up with brand new kind of quantitative high alpha ideas. That's what you want your research to do. And it's then they kind of take what I would call table stakes jobs, speeding up and picking up. Now, how that links to hiring, I mean, I think this is one we're all still trying to figure out, right? Because those very junior level graduate type roles where you would spend more time on the kind of BAU implementation, etc. Like it's kind of being squeezed out by these tools. And we're automating a lot of processes that used to require more humans. So it's a great question. Like how do you really interview, but for these skills? And I think it's, again, on the optimistic side.

The people coming in now have grown up and been more involved with these with the kind of frontier models. Maybe they are more literate than

kind of all of us. And so they have that advantage. I think the big disadvantage is that just. Pemptation at every moment, the answer to the problem you want to solve, especially when you're a junior, is just one button click away. Right. So whereas you and I, we had to go and dig out the papers, we had to read the theory, we maybe did some examples ourselves. I mean, the actuarial exam, you literally hand write out like Bruin theory you were talking about last week, you're writing out all of the maths behind that. The new generation just won't never have to do that.

You have to have some serious discipline to go and dig the book out when you know you can get a perfect one page summary with a click. So I think you've got to test, give someone an AI generated piece, get them to go and critically analyze it. What's wrong with that? What would have I improved on that? Because it's just a fact of life now, like locking someone in a Wi-Fi-free computer-free box is not a good test of what working life looks like today. Right. Yeah. That's very true. That's very true. You know, I don't know about you, but I actually often get the question during sort of more due diligence type meetings and people might ask, well, does all your researchers, do they do their own programming?

And of course the answer on so far has always been yes of course they do that part of what we want them to be able to do and they need to review each other's code and all that stuff right and. And I guess soon we'll just say no they don't they don't know how to program. We'll see what face they put up the investors if that's. You know a confidence face or a less confident face but only time will tell. Now. Before we move to the paper i want to make sure i think there might be more things do you want to link to more things before we dive into the first paper or because we're going to talk about the new paper that you co-wrote with a few colleagues joneson smith and chris pie i think it was but which is on on portable alpha but is there something more you wanted to bring up before we move into that harry.

Yeah, I think we can jump into the paper maybe just to set the scene a little bit because I guess a couple of the things that the paper really dives into you sort of discussed and I mean last week we before and if I go back to as well Rob Prokay and Kate Katie You mentioned some of the topics there as well so I think a good way to lead in it is to start with the cash efficiency of

Trendfollow, because that really builds into insupportable alpha. I mean, all of your audience already know this, but Trend is running on futures. You don't need to put up a hundred dollars of capital to get your hundred dollars of exposure. Now, this unlocks quite a few interesting features as an investor. If I only need to use 20 or 25 to get my 100 of trend following gives me a few new levers and things that I can do with the freedom of cash, right? So one common thing I hear is I like my portfolio. I've been happy with it. Like, I'm going to have to really kind of sell something down to free up some space for trend. And because of that capital efficiency, well, you don't have to sell very much or if anything at all.

Right, so you can get an allocation, you can get the potential benefits, the defensiveness, the convexity, the things that all your audience know very well for very, very little capital upfront. Now, the next thing I want to mention, and again, it links to the paper that Rob and Katie discussed was on. It's cash-efficient, right? You're getting this big benefit from directionally following markets. And in the paper that they discussed, they said, well, what is the additional benefit of the cross-sectional trend models? Could I replace cross-sectional trend models with, for example, carry? And I think the paper goes into, well, what other alpha sources might you want to put alongside trend? Now, because it's so cash-efficient, you're not having to stump up the full amount of dollars, that frees up cash to go and, I would argue, keep the cross-sectional parts.

lower correlation, it still had positive sharp, it was still additive, and add additional alpha models on top. So in this sense, you can sort of have your cake and eat it too, right? You're getting the full benefit of your trend allocation. You can fund it with, let's say, your 20 to 25 dollars. You can use that freed up cash to add some, the extra alpha in there. So that's one way of looking at the problem, right? Imagine, again, You're not getting your cash flows back as fast as you want from your private allocations. Maybe you have less equity there for or less cash available to invest. Pen following is potentially a nice way to get access because you don't need all of that cash back to get the full allocation.

The way this then links into the paper is essentially the paper is fully focused on portable alpha. So we've talked about a few ways that you could

maybe take the cash and invest in other alphas. You could just simply use less cash to allocate to trend. What the paper is doing is it is jumping into the portable alpha topic and saying, well, what I'm going to do here is I'm going to sell essentially no equity exposure. I'm going to keep my portfolio exactly as it was. And what I'm going to do is I'm going to take a portion of my equities, let's say I own some passive S&P, and I'm going to hold that via a future or a swap, and I'm going to port my trend following or my other allocation on top. Now, before we kind of dive in, the reason that we wrote this, and Jonathan and Chris were superb as well in contributing, a lot of the analysis and some on the structuring as well to this paper. But the reason we wrote this is that there's tons of introductory material on portable alpha now. If you go and search, you'll be able to find how should you do it? Should you use turnkey? Should you do it yourself? What are the risks of combining over 100% exposure? There's plenty of introductory material.

But what I found sitting in meetings and listening to people asking questions about portable alpha was the kind of next level down what I would call the sort of second and third derivative just wasn't covered very well. So if you go online and you search, for example, how should I rebalance portable alpha or how much cash should I hold in a portable alpha structure? Like you don't find great answers and you don't really find much, much analysis on there. That was really the impetus to go out and write this. Let's use some simple examples to make those points and bring them home. We published it. this week and you're happy to dive in. But if there's any questions up front, please go ahead. No, I mean, I think you're absolutely right. I mean, I think it's an important point that because I remember first coming across portable alpha back in the mid to late 90s talking about it. At the time, the firm I worked for, we never launched anything.

But the concepts were there. And then for me, I think really it came to the forefront again when Eric Crittenden and his group of people launched the standpoint product in, I think, 2020, just before COVID, if I'm not mistaken. That's really where it came back into light and say, yeah, this is interesting because to be very frank, There is this definitely a slide that has been sitting in my slide deck for for many years and that is just to visualize for people. What happens when you combine trend and equities because a lot of people think well if I put 50% here and 50% there I'll probably get an

average of the two but actually you get almost the sum of the two return stream which is the crazy thing. And so obviously it's become as you say it's become popular again many people have.

Launch products Korean his friends have really made it very public with the return stacking concept so that's another way of thinking about it so they done a great job in that and a big success in raising assets for it so. So yeah it's an important point I also have some push back that I get in meetings which I'll save and ask you a little bit later. But what I really loved about your paper is the fact that it actually does ask questions that even when I saw it, I thought, yeah, we never really talk about these topics. We should because they are really, really important. So from my point, I would just love for you to go through these questions that you so eloquently ask and answer.

Because I think people will be and obviously people should go and download the paper and read it, of course. But they're truly important in the context of this conversation without a doubt. Yeah, thanks so much, Niels. And it really was to answer some of those questions. Actually, I think we could... When I first kicked off doing some of this work, we thought, oh, this would be a week and it'll be so fast. And as you do it, you realize just how many questions you need to answer to build a... a resilient structure. So, yeah, if we dive in firstly on the title, I know last time I was on with Haiti and we were sort of touting writing a portable alpha paper. I think the working title was something like covariance of alpha and beta in stressed markets, some horrible. So, fortunately, we have a very good editorial and marketing.

team here and that the title is now Portable Alpha, Ask the Hard Questions, which I think is a much more clickable title than what we were, than what we had penciled originally. Yeah, sometimes it's good when you just say it actually does exactly what it says on the tin and that's kind of what this title illustrates. Yeah, absolutely. And so if we dive in, what will we really try to answer here? is a few things that continuously come up in portable alpha. Now, we know that in 2008, many of these structures failed. And the lines down which those structures failed, those risks still exist. So not enough

liquidity, correlations between your alpha and your beta, those things can still blow up and hurt you today.

Very much not saying, oh, don't worry about it. Like the world is different to 2008. It's very much saying, well, how can we build resilient structures that would survive docs like the GFC, like the dot-combust and like COVID? Now, the first thing I would pull out is essentially what we found on cash buffers in these strategies. Now, if you think of a portable alpha structure, You're buying your beta. Let's use S&P 500 as an example. You're then buying an alpha. And now throughout the paper, we use SG trend, obviously as a trend podcast. Slightly biased here, but there we are. But you could pick multi-strategy, you could pick other alphas in there. And we've used trend in the analysis, but internally we've done this analysis across many different alpha structures.

And so you're buying 100% of your beta and then you are layering the alpha on top. So what you're aiming to do is replicate the passive index. You're aiming to deliver the S&P to your investors plus whatever alpha you can get on top. Now, obviously, when you put your S&P onto a derivative, onto a future or a swap, you're no longer putting up the full 100 to get your 100 exposure. Now, you can put as little as 10 in to get your 100. Right now, that frees up 90 to go and buy lots and lots of alpha. The problem is in a stress scenario where your beta starts to sell, well, your counterparties are going to start calling you and saying, hey, your position is losing. You need to send us a small margin. Now, if you've only supported your S&P with 10 of capital and the S&P falls by 10%, all of your margin is gone, right? And say you need some unencumbered cash. So it's a given. You have to have some unencumbered cash in the structure. Now, what I think is really, really interesting is you might then say, well, okay, I'll have an extra 10 of unencumbered cash. That should keep me nice and safe and like I can wear there a pretty big drawdown. Now, that kind of 20% unencumbered cash, if you hold it and you stimulate this, S&P plus trend portfolio through the last 26, 27 years, that gets fully drawn down three times. So you get fully drawn down in the GFC, in the dot com and in COVID. So your structure will function fine most of the time. And then in these big left tail events, well, you're most likely going to have to close out some of your beta position. Now, this is a really good question as well because Is that bad? Now if you

are an allocator and you have benchmarks and from one day you are 100% long the S&P like your benchmark and then the next day you are 20% long the S&P because you've had to sell along that position, you have now massive tracking error versus your benchmark. The second point is that over the long run equities have tended to go up and after big crises they've tended to rebound.

And so one part that we look at is if you were selling in the early part of 2009 at a low in your equity component and you're knocked out of that position, you miss 29, 30% of upside. So it can be a pretty bad outcome. It's certainly an outcome as risk managers. We do not want to happen. We want it to be a very, very left tail, very far away probability. So what we look at is we say, well, what is a cash buffer that starts to really make it very, very unlikely this will happen. Now, I say unlikely, not impossible. Right. And in the paper, we end up with essentially 40% cash to support your beta. So that's 10% margin, 30% unencumbered. And our approximate calculations mean that that will get shut out as in you'll have to de-gear that beta about once in every one to 200 years.

Right, so it's not impossible. Essentially what we're saying is if you look at market data back through time, look at live data, it survives. But obviously you've only had one run of markets. Again, you talked about this at length last week. You've only had one path of the truth that actually happened. Now we know you could get bigger drawdowns than that in the full distribution. And so by running Monte Carlo, we're able to simply sort of severe drawdowns. And we think that kind of one in 100, one in 200 year sort of timeframe is reasonable, but it's very, very unlikely manageable if it happens. You'll have less equity exposure. You can regear it afterwards. But that's sort of where we looked at that. Now, what I think is the interesting finding for listeners is if you're looking at, say, a 20% buffer, suddenly you're massively increasing the chance that you get knocked out.

Some people might be happy, might be fine with that. They might have callable capital available. They might have a credit facility. They might be happy to get on the phone with counterparties and post that margin at short notice. Now, a lot of investors that we work with do not want that. They want us to take on the operational complexity, the cash flow complexity, the

risk management, etc. They do not want to be getting margin calls on a Monday. Please send cash by end of day. So that was one finding on the cash buffers. Now, the next bit that we essentially were also looking at, well, I mean, firstly, you can look at the alphas that you hold, like, why just can't make a nice alpha in Port of Alfa? Well, it's super liquid, right? If you have something that's like weekly dealing or monthly dealing, you can recap slides quickly. So it makes trend day very, very nice.

candidate for it, but you can, of course, use others. And the next one we really dug into, which, again, it started coming up into in my meetings, was, well, how should I rebalance my Alfa and my Visa? Now, at first, this sounds like the most simple question. Well, obviously, just pick Buffy or pick Quarterly. Can't be that difficult, right? That's how I rebalance my alt portfolios, however I balance my equities of bonds. And then I would say, okay, so we're going to pick monthly rebalancing. Now, at year end, do you want the return of the S&P in your beta component? And they would say yes. And I would say, well, as soon as you select a monthly rebalance, you no longer replicate the S&P. And this is actually quite interesting new news to a lot of people. And when I first was looking at this, I was like, oh yeah, this makes complete sense, right? If you own the S&P, You don't buy it and sell it every month. You just own the index, right? If it goes to 110 and then it makes another 10%, it compounds on itself. If it goes to 110 and you sell 10 of it to rebalance, you're no longer tracking the index. When you get to year-end, you're going to get a surprise that, oh, my portable alpha beta component has lots of tracking error versus the index. Then we've run a number of different rebalancing rules. If you want to crack the index as close as possible, you should actually buy and hold that index. So maybe do an annual rebounds or even longer. Just hold it and hold your alpha. Now, in the paper, we say we wouldn't do this ourselves, right? Because as you buy and hold, you're going to slowly get this drift. You might own too much equities or too little. If other investors come in, how do you get them to the nav? Do you give them this non-one-to-one mix that they're looking for?

So we think, essentially, if you can let that beta replicate closely, so rebalance a little bit less, but have some sensible tolerance. So let's say not going to rebalance, but if my S&P and SG trend goes more than 10% out of whack, I will rebalance. Now, it's all very good then talking through this.

We put some numbers on it to look at specific episodes. So something like 2008, it pays to rebalance very slowly because your equities are falling, your trend allocation is rising. Every time you rebalance, you rebuy, you increase your equity allocation into a continually falling equity market. You sell trend that's rising, essentially all rules other than a buy and hold underperform in 2008. Now, that's just one example, right? It is one that everyone quotes. Like everyone who is in this business will always talk about 2008 and rebalance it.

But what about over the long run? And over the long run, we found that in performance terms, it kind of doesn't matter. So whether you do monthly, quarterly, annually, whether you do 5%, 10% tolerance, it doesn't matter over the very long term. But you will get periods where it really matters, like 2008, potentially like COVID. And so all we wanted to do is just to highlight, look, it's a good question to ask is just have a look at the paths. Are there any paths that you really wouldn't like in there? And therefore, that might push you towards a certain rebalancing rule. So we look at that. Again, it comes up in so many meetings. We wanted to put something out. Again, we've internally done it with other assets, but here we're doing trend. And the final bit of the paper, which again, if you Google, should I use what's all futures in portable alpha? Oh, yeah. Not a ton of information, right? No, that's a good question.

Yeah, so that's another one that we looked at. So he ran some, so essentially, they're kind of no free lunch, no arbitrage assumption, which it obviously holds in a market as big as this. But I just wanted to see it proved, right? Because you come to this and you say, hang on, though, my counterparty bank is going to charge me 40 basis points over Doffer. Why don't I just go and buy the future where I don't see the 40 basis points over. Whereas there's an explicit cost in the swap, it's implicit in the future. And if you run this over time, it washes out, right? You don't get a structural performance advantage. So what we describe there is essentially it comes down to operational preferences. Internally, we run futures, we run swaps, we make the case that it's probably down to investor preference.

And there's some operational considerations, obviously futures. You don't have the counterparty risk. That might be a consideration for some in a

structure like this as well. So we dive into that. And that's kind of where we end. And really, the conclusion is that there isn't a completely right or wrong answer for any of these. But you should be talking with your providers about these questions. You should understand what will happen if you draw down 20% in a month. Like it's important to know ahead of time. Yeah. Yeah. No, no. I mean, that's great. And I also think it's important because I mean, obviously you do touch on correlation. We obviously talked about it initially about why trend is such a great candidate. But as these strategies become more and more popular, we have seen some other combinations that may look great in a backtest, but.

where there might be some risks that hasn't been exposed in the last couple of decades, partly because of correlations, partly because of other things, but I think you did a great job in touching on all of that. Now, in consideration of time, I think we have maybe another 10, 12 minutes left, and I know there's another paper that we came across. That you really liked and and it's about the mathematics of the turtle trading is the name of the paper written by Nicholas Paulson and Vadim. Circle of and I would not mind if you could give us a little bit of a flavor of what's in the paper why you liked it. So maybe we don't have time for a full deep dive, deconstruction of the paper, but we kind of focus on some of the highlights and then people can go and read the paper. I'm sure I'm going to link to it in one of my weekly emails, but I know you found it very interesting. So why don't you talk a little bit about that, Harry? Yeah, absolutely. And thank you for flagging this one, Niels.

stupid breed. I really enjoyed it. It did absolutely take me back n-class years now to my actuarial days. A lot of the concepts in there like you're tested on in the fairly abstract and it's just lovely to kind of see them all unified here. But essentially what the paper is doing is it's saying the turtle rules that were applied in the 80s and then published in the early 2000s. These were kind of rules of thumb, hand-calculable pen and paper, easy to do in a few minutes, approximations of later decision theory mathematics that was kind of published in academia. And what the paper is doing is it is linking those original rules from the 80s to academic papers starting from the 50s, but all the way up to now. And so it's a beautiful combination of them.

them history, some financial history, behavioural economics. I know you've talked about the turtles and the behavioural side, but here it's really relating that to what was later published in a decision theory, probability theory by authors. So when I read this, I also opened the original turtle rules and read those original turtle rules as well. And obviously the story there, A group of people with limited or no trading experience, given a set of rules, follow these rules and you will be a profitable trader. And we know some of those came out and were very successful. But I mean, this isn't the area to touch on your much bigger expert on that area than me, Neil. So let's actually jump into the concepts that they bring out in the paper.

And the first one, which is a favorite of the audience and all of us practitioners, is estimating volatility. Though the turtles have a property N, which is essentially the volatility of a market. And they use a range-based estimator. Very simple to update each day. It's essentially like an exponential decay, half-life 14. The people calculating this with pen and paper were simply looking at open close, high low. and getting some volatility estimate for each market. So that's what we would today see as our vol estimate in crude or our vol estimate in treasuries. Now, what's interesting is the paper then links that to published research, for example, Parkinson, who says, look, this is actually an efficient way of estimating vol. Their kind of pen and paper calcs that they were doing on the fly is then backed up in academia as like, yeah, this is a legitimate way to estimate vol.

Now, where I think it starts to get more interesting is on, well, you now have all estimates for your markets. How do you position size? Like, how do I take that information and risk weight my positions? And so the original turtles, they had this idea of a unit, such that one unit of each commodity or financial asset they were trading would be equivalent in risk terms. You can imagine a unit of crude oil, and a unit of crude oil versus a unit of 10-year US Treasury. They're equal in risk terms, even if you have to buy five times the dollar amount of the Treasury. Really, what this is, is this is volatility-based position sizing. It's risk-based position sizing done in the 80s by the Turtles.

Where this then starts and what the paper does is it then relates this to a lot of the academics around these position sizes, optimal bet sizing. And so they bring in the Kelly criteria, which again, you've discussed a great length. They quote Ed Thorpe and for your listeners, if anyone hasn't read his The Man for All Markets, I mean, that is probably, I would say, the best description of how to implement this in practice. And essentially, it's saying that the optimal bet that you want to take to kind of compound and grow wealth over the time, you need to take some account for the probability of going bust, right? Like the ruin theory that you've discussed. And as you increase your bet size, you increase your expected return, but you're always adding this tail that you might hit. I mean, one of those actuaries would call ruin. I think ruin theory is very applicable here.

And what the turtles did is they had stops in based on the risk of that position. Though this quantity n, which is your estimate of vol at 2n, you take your position away. You stop out of that position. And what the paper is saying is, well, look, this 2n equates to 2% of your total equity in your book. That stop is set. You can't lose more than 2% of your equity. And it links it to... for the Kelly criteria, an optimal Kelly. Now, what the authors describe and what they actually show, and they do attest with their own data later on, is this is quite far on the kind of Kelly hill. You have some optimal Kelly, which is under no uncertainty, and that maximizes your long run compound growth. Now, any prudent manager will be on the left side of that hill.

Right, you are taking a little bit of less of the upside for a much lower probability of ruin. And what the paper shows is that these turtle traders back in the 80s were running a version of that, a pretty prudent version of that, where their risk of ruin was much, much reduced. So again, for listeners, I think that the Ed Thought Book describes it fantastically well. The paper also kind of links all these concepts really nicely. Now, the next thing they look at, and again, very, very interesting for like a modern day trend follower, is in that position sizing, obviously it's not just volatility that impacts that, right? Like, as we all know, we should be considering things like liquidity. We should especially be considering the correlation between these markets. So any modern risk?

risk position sizing will also include obviously information on the correlation between markets. Now, remember, back in the 80s, these traders were using pen and paper to calculate, right? They weren't inverting covariance matrices, and they had to have rules of thumb. And so the way they dealt with correlations was not in the actual construction, but it was on position size caps. So correlated markets They would cap. And again, what the paper does is it essentially just links this to academics, more modern decision and portfolio construction theory, which is just, it's beautiful that they were kind of doing it in the 80s. And the final point I would touch on, which was the, I mean, there's a few more, just in the interest of time, they also had drawed emeralds, right?

Many managers will have this, certainly multi-strategy pods. You know you have these very, very strict drawdown rules. But anyone really interested in the maths of Kelly and drawdown control, you know under your Kelly, if you are left at the top of the hill, you shouldn't go into ruin. You've got a decent probability of avoiding ruin. The issue is, if you're down 80% in your strategy, You still know it's going to work, right? If it's your strategy and you're following the rules and you are like indestructible in your confidence, you don't suffer from human biases, you implement that like you should be fine. The issue is, and like in the turtle's case obviously they're being given a notional book, someone else's equity in a manager's case, you are running allocator's money, you are running the money of pension scheme participants, teachers, etc. You cannot stomach an 80% drawdown. So what is an optimum rule for mitigating those drawdowns? And so the turtles back in the 80s, I'm just trying to recall, I think they used a, they lost 10% in the market, then they would cut their equity by 20%.

So you're essentially approaching a limit where you essentially can't lose any more at some point. Now, what the paper does is it links this to the later published work of Grossman and Joe in the 90s on optimal drawdown control metrics. And again, it's like a beautiful link, right? Again, these hand count rules back in the 80s that were given to the kind of regular regular people. You can actually express that in in maths very very nicely and people do that that later. So yeah, I really enjoyed reading it. It's a bit of a history lesson as well as some just some very nice times there again. You're the you're the probably the expert on the on the history more than

me, but I did enjoy this one. That's very kind of you to to to take us through that and the links to sort of research that's been published later on. But you're right. I mean, I think there was a time where people thought maybe that because the turtles, they were not the first trend followers. You know, there were trend followers in the 70s as well. But they became the most known in the sense because of the books that was written about them and the narrative. We know how strong narratives can be.

And I think there was certainly a period of time where maybe people mistook the clarity of rules as being something that was simple, but actually the genius was not so much in buying a hundred day breakout. The genius was, as you point out, in the way they approach risk. And I also think that sometimes with trend following and CTAs in general, I think it's it's not understood well enough that we are probably risk managers first and foremost actually the returns that we generate is not something we control but we do control. To large extent the risk we take so I've always felt that we as an industry we came from it from a risk management first and foremost. So yeah it's sort of building a system that takes small losses and scales intelligently and remains intact when the.

You know the outsized opportunity finally arrives. That's essentially what what we've been doing for the past five six decades in our industry. And it's nice to see these things being continued talked about and written about in all of that. So really, really grateful for you doing all of this heavy lifting today. Harry it was. really good and also getting your additional perspective and thoughts and let me actually go back one step before we round up in the next few minutes. Because there's one thing I forgot to ask you about the portable alpha. I'm so excited about hearing your run down of the portable alpha paper. I do come across people who are interested in the concept, but where they feel a little bit worried about adding leverage, which is essentially what the product does. It combines two return streams and it's more than 100 in total. It becomes maybe 150, maybe 200. So in their view, it's adding leverage and it's replacing two unlevered investments, so to speak.

I asked the same question of rich and Dave last week, but I love to hear your kind of brief perspective and that is, and I don't want to make it too

much of a leading question, but I think it's difficult for me not to. Why should people perhaps be less worried about the fact that you're stacking these two return streams on top of each other? You're putting the alpha on top of the beta, so it looks like it's a leverage product. Why should people be less concerned about that when we're talking about CTA or trend being the alpha generator? Yeah, Niels, it's a great question. It's like the leverage does just push up a bit of a kind of a version barrier in some investors. And I mean, rightly so in some areas of the market, right? I think Dave and Richard last.

We were talking about the kind of three times level ETF and the asymmetry of the return profile, like leverage there is scary. The way I see leverage and the way I consider it is that it is a tool. It's a portfolio construction tool. It's an investment management tool. Before linking it to Trend and Port Laofar, let's just take a portfolio of equities and bonds. If you start with a portfolio of 50% equities and 50% bond, What you find is that pretty much all the risk of that book is coming from the equity portion, right? 85% of the risk is equities, 15% is bonds, even though on paper it says 50-50. Now, the optimal kind of risk weight or the optimal sharp ratio weights of those assets is going to be more bonds than equities to get more risk balance, right? It might be 30% equities, 70% bonds. Now, nobody likes that portfolio because the return's gone down, right? Sure.

bonds are less risky, they have lower return. So if you allow leverage in that portfolio, maybe you hold your 70 bonds, you're still able to hold 50, 60, 70% of equities. You get a very risk balance portfolio, you get a higher sharp, and then you would buy both by having the 100 constraint. Now, in portable alpha, the alpha that you port is very important to your overall outcomes and your probability of having a bad experience. So in your alpha component, you're layering alpha onto 100% beta. So you already have your 100% S&P. You've got your 100% equities. Now you're introducing an additional alpha driver on top. So when you look at that alpha driver, what alpha driver do you want? You almost certainly don't want to be introducing more beta, right? That's a risk in the same direction. That is really pushing out your volatility. It's pushing out your...

potential of a big left-tail loss. We also need to really investigate. It's not just that it's uncorrelated, but that it's uncorrelated in the left-tail. And the reason this is important is that plenty of alpha strategies that are kind of liquidity provision, and they look very, very good in normal times, but have a big left-tail. Like short vol is the most extreme of those. You could even think of something like merger arbitrage. treasury basis trade, something where if you have a very, very big shock, that little rule that was making you money, I mean, all the deals might break in Mergerab or in the basis trade, the liquidity dries up and you get really hit because you know the liquidity you need.

A good thing to do is actually look at the returns of your Alpha source, look at your returns of your Beta source, not just the whole data set, but the worst 5% of Beta returns, like the worst equity periods and check that you still have no correlation. Now, why does Trends sit nicely in there is because in those worst periods, Trends actually tends to have a negative correlation to equities. In your real tough times for your structure, where your equities are really getting hit over like a three, four month period, you're down a decent amount there. And historically has had a very low negative correlation. So it makes a very good compliment. And back to your original question on the leverage, it's extremely cash efficient. You're not actually deploying many dollars to get that exposure. And also, when there's a big turning point in markets, what happens? Falling markets goes right up.

your trend positions come right down. Like we just discussed in the turtle paper, you scale with volatility, risk has gone up, your position size has actually come down. So, where people are worried about margin calls, you'd be especially worried about this in like a liquidity provision strategy, like for basis trade or other structures like that, trend is actually reducing positions. It needs less margin. That's a great point on leverage. I see it as a tool and certainly a tool to be well understood. But if it's increasing diversification, then my view is that it can be used to reduce risk. Fantastic. Good stuff. Harry, really, really wonderful. Thorin enjoyed it. I know the audience will as well. And for those of you listening in, if you did, show Harry how much you appreciate all the preparation he did.

I go into your favorite podcast platform and leave a rating and review because it really does help more people discover these conversations and concepts. And of course, as mentioned, make sure you go to the inside page on man's website and find the paper that we discussed today. And there are lots of other great resources on that. So that will give you enough to read until you join us next week where I will be joined by Andrew beer. and Tom Robel where we will continue our conversations about many of these topics. If you do have a question feel free to suggest your topic and send it over to me and I'll do my best. The email is info at toptradersonplug.com which actually happens to be also the email address you can do if you leave a really nice review. As mentioned in last week's conversation where you can actually win.

a TTU merch or in terms of a vest, a really nice vest, I have to say. You might have seen some people wearing it on the show. I need to make sure you get one as well, Harry, of course. But anyways, there is some instruction in last week's episode at the very end. So go and check it out because at the end of this month, we will find a winner. And some of the reviews we've had so far are actually very good. So you need to up the game if you want to be Considered for that anyways from Harry and me. Thanks ever so much for listening We look forward to being back with you next week and until next time as usual take care of yourself and take care of each other Thanks for listening to top traders unplugged If you feel you learned something of value from today's episode, the best way to stay updated is to go on over to your favorite podcast platform and follow the show so that you'll be sure to get all the new episodes as they're released. We have some amazing guests lined up for you and to ensure our show continues to grow, please leave us an honest rating and review. It only takes a minute and it's the best way to show us you love the podcast. We'll see you next time on Top Traders Unplugged.

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